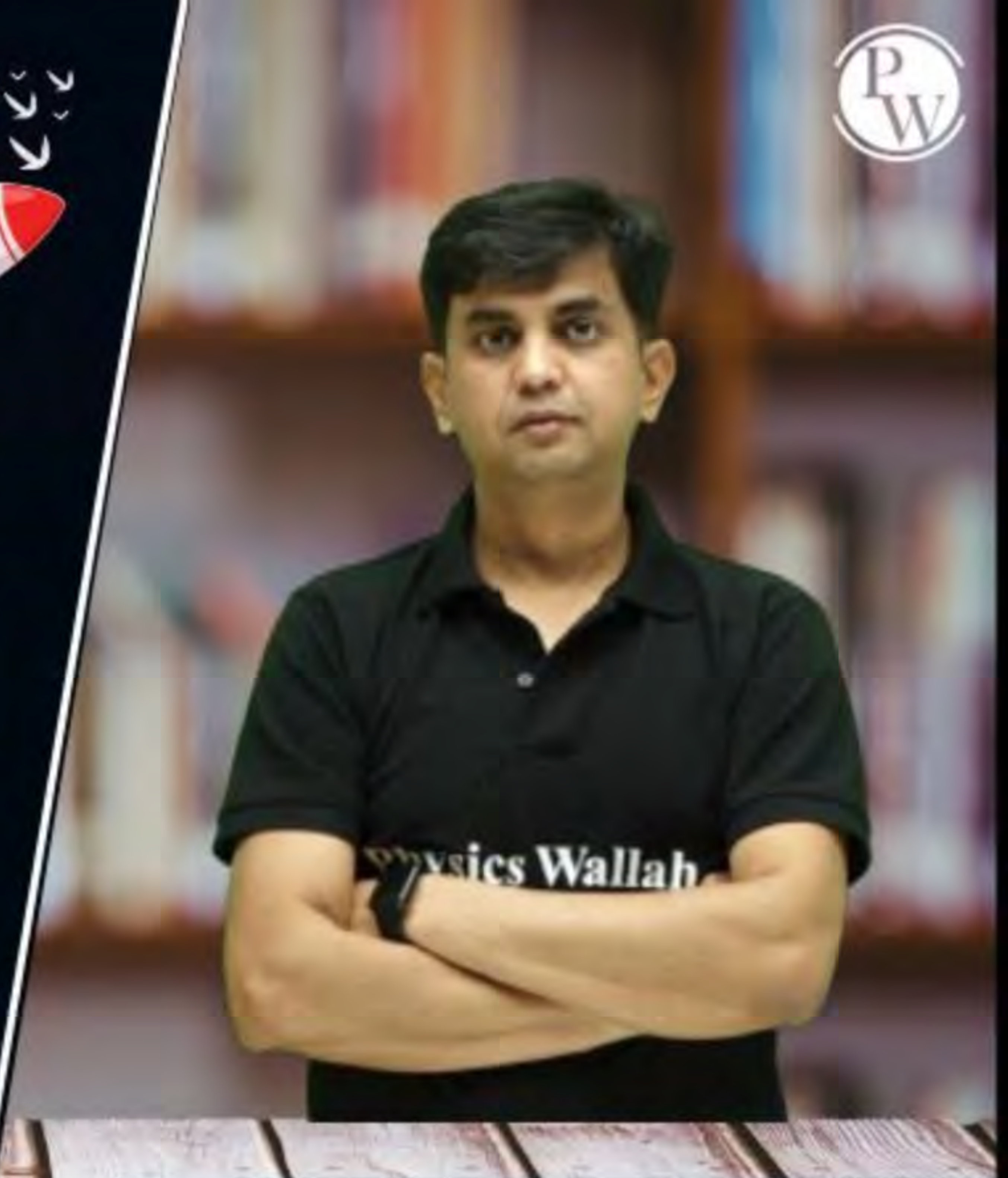
CS & IT ENGINEERING

Computer Network

Routing

Lecture No. – 01



By - Abhishek Sir

Recap of Previous Lecture

Topic Forwarding

Topic ARP

Topics to be Covered

Topic Routing

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
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[MCQ] [GATE-CS-2006] [2 Mark]

#Q. Two computers C1 and C2 are configured as follows. C1 has IP address "203 . 197 . 2 . 53" and netmask "255 . 255 . 128 . 0". C2 has IP address "203 . 197 . 75 . 201" and netmask "255 . 255 . 192 . 0". Which one of the following statements is true ?

(A) C1 and C2 both assume they are on the same network

(B) C2 assumes C1 is on same network, but C1 assumes C2 is on a different network

(C) C1 assumes C2 is on same network, but C2 assumes C1 is on a different network

(D) C1 and C2 both assume they are on different networks

$\boxed{\text{Ans : C}}$

Host C_1

IP_1	:	203 . 197 . 2 . 53
$Netmask_1$	:	255 . 255 . 128 . 0
$Net. Add._1$	:	203 . 197 . 0 . 0

IP_2	:	203 . 197 . 75 . 201
$Netmask_1$	:	255 . 255 . 128 . 0
Net. Add.	:	203 . 197 . 0 . 0

Same

Host C_2

IP_2	:	203 . 197 . 75 . 201
$Netmask_2$	:	255 . 255 . 192 . 0
$Net. Add._2$	:	203 . 197 . 64 . 0

IP_1	:	203 . 197 . 2 . 53
$Netmask_2$	:	255 . 255 . 192 . 0
Net. Add.	:	203 . 197 . 0 . 0

diff-
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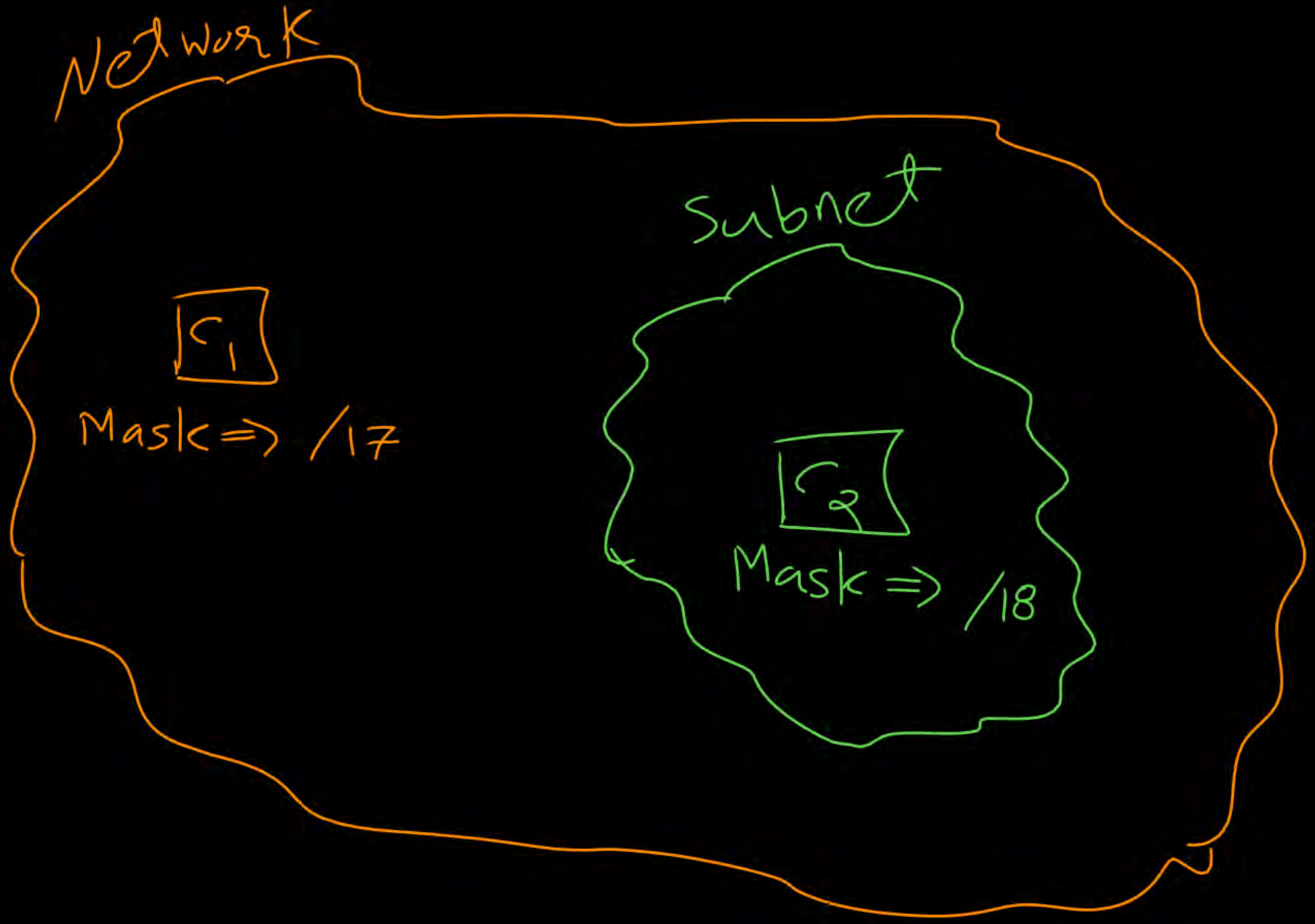

Network
Subnet
C_1
Mask ⇒ /17
C_2
Mask ⇒ /18

[MCQ] [GATE-IT-2008] [2 Mark]

#Q. Host X has IP address 192 . 168 . 1 . 97 and is connected through two routers R1 and R2 to another host Y with IP address 192 . 168 . 1 . 80, Router R1 has IP addresses 192 . 168 . 1 . 135 and 192 . 168 . 1 . 110, R2 has IP addresses 192 . 168 . 1 . 67 and 192 . 168 . 1 . 155, the netmask used in the network is 255 . 255 . 255 . 224;

1 Which IP address should X configure its gateway as?

A 192 . 168 . 1 . 67

B 192 . 168 . 1 . 110

C 192 . 168 . 1 . 135

D 192 . 168 . 1 . 155

Ans: B

Netmask = 255 . 255 . 255 . 224

97 = 01100001
110 = 01101110

135 = 10000111
155 = 10011011

67 = 01000011
80 = 01010000

[MCQ] [GATE-IT-2008] [2 Mark]

#Q. Given the information in previous question, how many distinct subnets are guaranteed to already exist in the network?

2

A 1

B 2

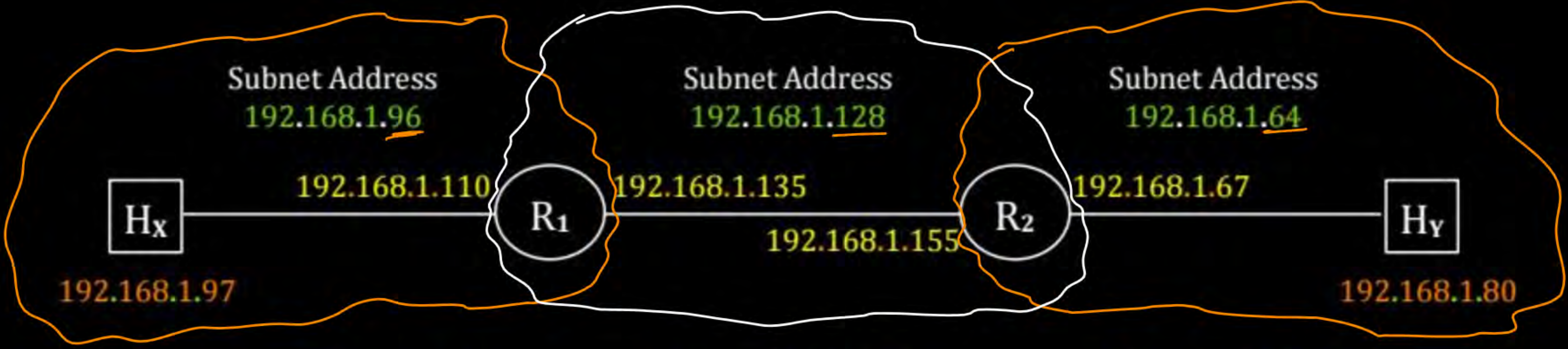
C 3

D 6

$\boxed{\text{Ans: C}}$

001
101
110 ?

Netmask = 255 . 255 . 255 . 224



97 = 01100001
110 = 01101110

96 = 01100000

135 = 10000111
155 = 10011011

128 = 10000000

67 = 01000011
80 = 01010000

64 = 01000000

[MSQ] [GATE-2024][1 Mark]

#Q. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

(A) The destination IP address is the IP address of R F

(B) The destination IP address is the IP address of Y T

(C) The destination MAC address is the MAC address of S

(D) The destination MAC address is the MAC address of Y F

$\boxed{\text{Ans: B}}$

Network Address : 192 . 168 . 0 . 0 / 16

With 8-bit subnetting

Source IP Address : 192 . 168 . 1 . 2 Hx

Destination IP Address : 192 . 168 . 2 . 2 Hy

IP Packet

→ Host X (source host) finds destination host IP (Host Y) belongs to different subnet

→ Host X uses ARP Protocol to find MAC Address of default gateway [192 . 168 . 1 . 1]

→ Host X send frame to Router [The frame encapsulates the IP datagram]

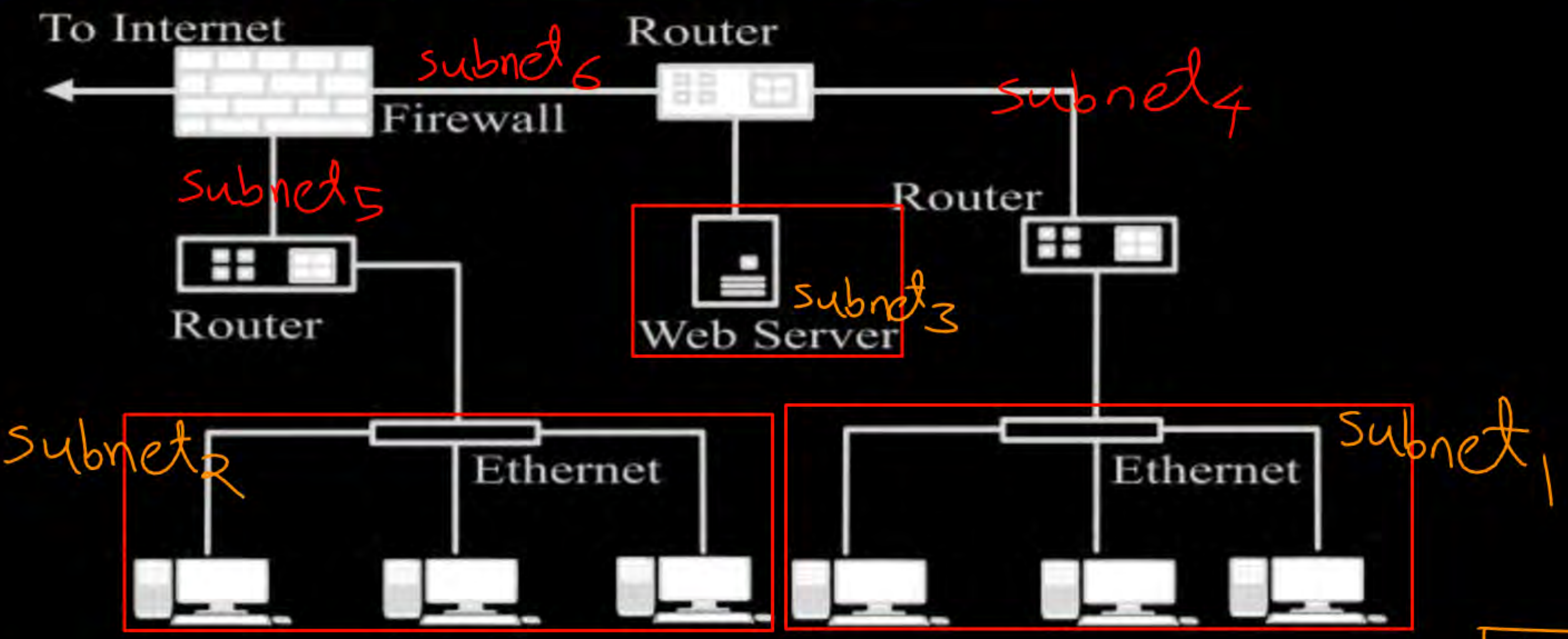
Source MAC Address : Host X MAC Address

Destination MAC Address : Router MAC Address

Frame[IP Packet]

[MCQ] [GATE-2022][1 Mark]

#Q. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.



What is the number of subnets inside the enterprise network?

(A) 3

(B) 12

(C) 6

(D) 8

Ans: C

[MCQ] [GATE-2019][2 Mark]

#Q. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

A. X sends an ARP request packet with broadcast IP address in its local subnet

B. X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X

C. X sends an ARP request packet with broadcast MAC address in its local subnet T

D. X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

$\boxed{\text{Ans : C}}$

Topic : Syllabus

Routing protocols : shortest path, flooding, distance vector and link state routing;

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* $\underbrace{\text{path vector Routing}}_{\text{Not in syllabus}}$

Topic : Forwarding

Data Plane :

-> Determine how IP packets are forwarded

[Using Router's Forwarding/Routing table]

-> Move packet from a router's input link to appropriate router's output link

Topic : Routing

Control Plane :

-> Determine how IP packets routed among routers

[Determine route taken by IP packets from source to destination]

-> Construction of Router's Forwarding/Routing table

[Using Routing algorithms/protocols]

Topic : Routing

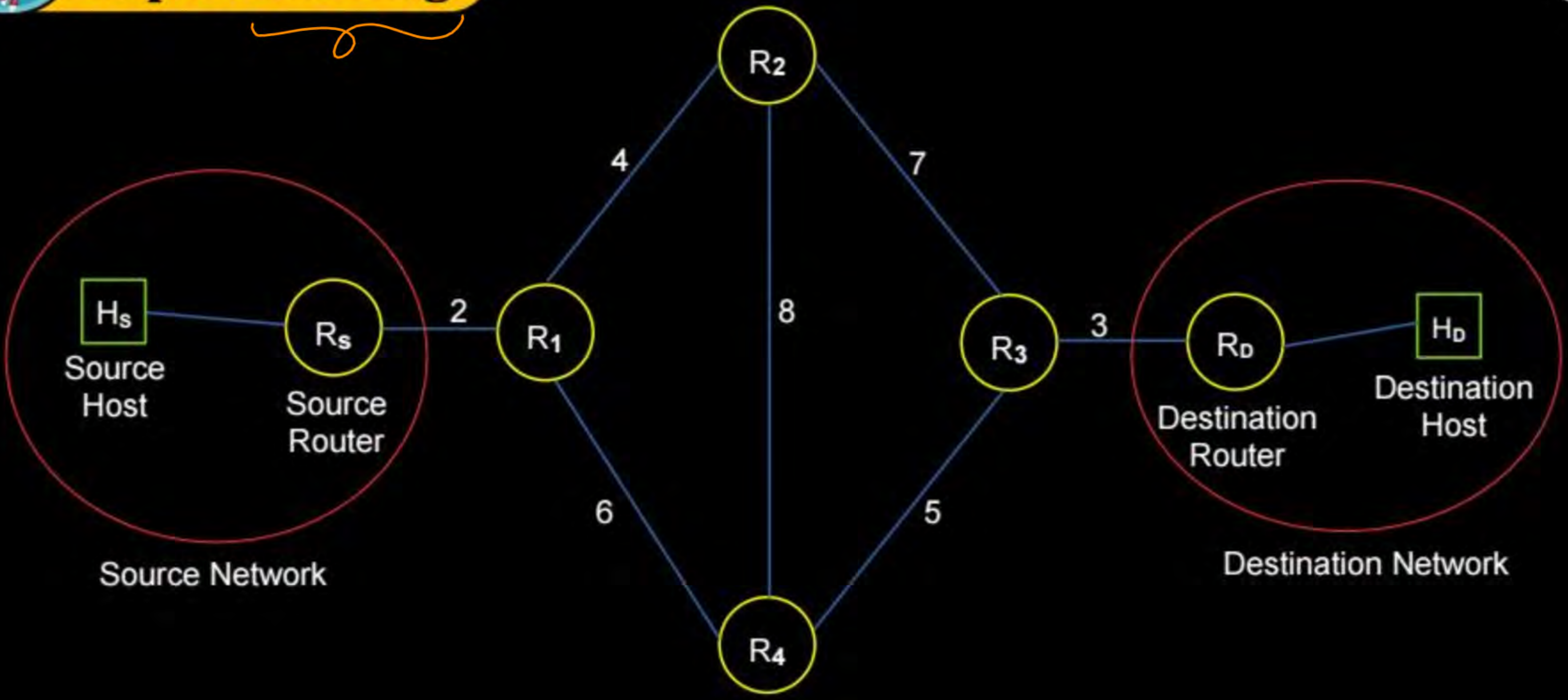
→ Identify optimal (best) path between Source and Destination.

→ Types of Routing :

1. Static Routing
   → Non-adaptive Routing ⟹ Mannual

2. Dynamic Routing
   → Adaptive Routing ⟹ No mannual (plug & play)

Topic : Routing



Topic : Link Cost

→ Metric to compare link :

1. Distance (Hop Count) → RIP - DV
2. Delay → OSPF - LS
3. Bandwidth

Cost $\propto$ Distance, Cost $\propto$ Delay, Cost $\propto \frac{1}{\text{Bandwidth}}$

Topic : Static Routing

→ Non-adaptive Routing

→ Types of Static Routing :

1. Shortest Path Algorithm
2. Flooding

Topic : Shortest Path Algorithm

=> Single-source shortest path algorithm

→ Dijkstra's algorithm ⟶ Greedy

→ Bellman–Ford algorithm ⟶ D. P.

=> All-pairs shortest path algorithm

→ Floyd–Warshall algorithm

Topic : Dijkstra's Algorithm

→ Dijkstra's Algorithm is divided into two steps :

1. Initialization
2. Iterative distance calculation

Topic : Dijkstra's Algorithm

Input

[Graph]
[Adjacency Matrix or List]

[Source]

Dijkstra Algorithm

Shortest path to all other nodes

$\Downarrow$

Forwarding (Routing) Table

Construction and dynamic Updation

$\Rightarrow$ Mannual

2 mins Summary

Topic Types of Routing

Topic ~~Link State Routing~~

THANK - YOU

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